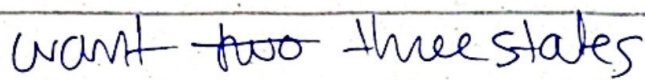


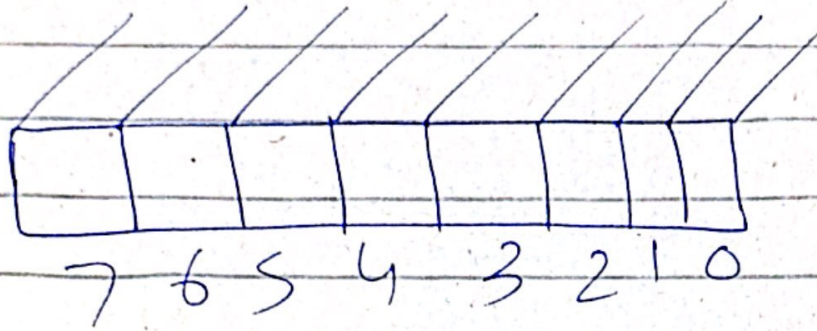
pull v pull $\hookleftarrow$ b
down .. up ..



Floating (only term ~~when~~ pull up
pull down used to drive the circuit)



Each 32-bit



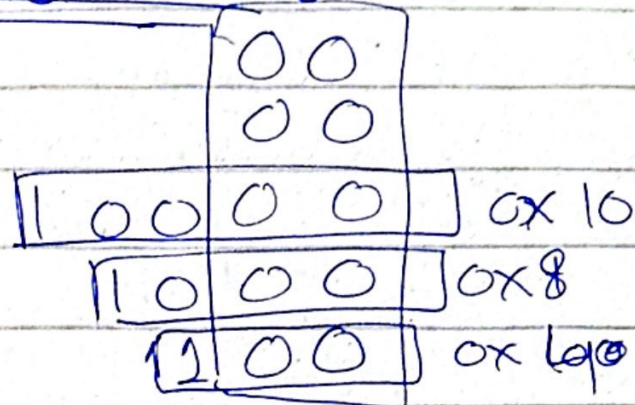
→ Every ~~32-bit~~ pin has a 32-bit register

0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	0
1	0	0	0

2nd 0th

always zero

for Data Register only:



9 8 7 6 5 4 3 2 1 0
 pin pin pin pin pin pin pin and pin
 7 6 5 4 3 2 address address

Access pin 0 and pin 3

11 0 0 $\Rightarrow$ 0xC $\Rightarrow$ pin 0 and pin 3

00 1 1 1 1 1 1 1 00
 3 F C

offset 0x3FC $\Rightarrow$ All pins
 $\rightarrow$ can manipulate using a single macro.